

Homework (Carolina Reaper)

- Q1.** A music festival has a maximum capacity. If x represents the number of attendees and y represents the total revenue, which correlation would you expect to see in the scatter plot?
- (A) Positive
(B) Negative
(C) None
(D) Cannot determine
(E) Inverse
- Q2.** A painter notices that the larger the canvas size (x), the more time (y) it takes to finish a painting. What type of correlation can be expected?
- (A) Positive
(B) Negative
(C) None
(D) Inverse
(E) Direct
- Q3.** In a rhythm pattern, if the number of beats (x) increases, the time to complete the pattern (y) also increases. What can be said about the correlation?
- (A) Positive
(B) Negative
(C) None
(D) Cannot determine
(E) Inverse
- Q4.** The frequency of a musical note (x) and its corresponding wavelength (y) have a correlation. Which of the following is most likely true?
- (A) Positive
(B) Negative
(C) None
(D) Inverse
(E) Direct
- Q5.** A line of best fit is drawn for the data points (1, 2), (2, 3), (3, 5). If the line of best fit is $y = mx + b$, what can be said about the value of m ?
- (A) $m > 1$
(B) $m < 1$
(C) $m = 1$
(D) $m = 0$
(E) Cannot determine
- Q6.** A scatter plot shows a strong positive correlation between the number of hours (x) a musician practices and the number of songs (y) they can play. If the line of best fit is $y = 2x - 1$, what is the predicted number of songs a musician can play after practicing for 4 hours?
- (A) 6
(B) 7
(C) 5
(D) 4
(E) 3
- Q7.** The data points (2, 4), (4, 6), (6, 8) are plotted on a scatter plot. Which of the following lines of best fit is most suitable?
- (A) $y = x + 2$
(B) $y = 2x - 2$
(C) $y = x$
(D) $y = 2x$
(E) $y = x - 2$
- Q8.** A composer notices that the longer the duration (x) of a song, the more frequently (y) it is played on the radio. If the line of best fit is $y = 0.5x + 1$, what is the predicted frequency of a song with a duration of 10 minutes?
- (A) 5
(B) 6
(C) 4
(D) 3
(E) 2

Open Ended Questions (FRQ)

- Q9.** The table below shows the relationship between the number of art pieces (x) created by an artist and the total revenue (y) generated. Create a scatter plot and draw a line of best fit. Use the line of best fit to predict the revenue generated when the artist creates 10 pieces. x : 2, 4, 6, 8 y : 100, 200, 350, 500
- Q10.** A musician notices that the more they practice (x), the better they perform (y) in concerts. The table below shows the relationship between practice hours and performance ratings. Create a scatter plot, identify the correlation, and draw a line of best fit. Use the line of best fit to predict the performance rating when the musician practices for 12 hours. x : 1, 2, 3, 4 y : 60, 75, 80, 92

Chef's Tips

- Tip 1: Encourage students to use graph paper for scatter plots.
- Tip 2: Remind students to label axes and title their graphs.
- Tip 3: Suggest using technology to create and analyze scatter plots.